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Electric sign device

Abstract

An electric sign device can include a backboard; first and second metal strips attached to the backboard; a plurality of LED assemblies, each LED assembly comprising: a board; first and second metal contacts attached to a first side of the board; first and second magnets attached to the first and second metal contacts, respectively; the first and second metal contacts contacting the first and second metal strips, respectively; a plurality of LEDs disposed on a second side of the board, the plurality of LEDs being electrically connected to the first and second metal contacts; and a carrier having one or more channels or apertures that receive the plurality of LEDs, the carrier being attached to the board; wherein the first and second magnets are magnetically attached to the first and second metal strips, respectively, so that the each LED assembly is magnetically attached to the backboard.

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Background/Summary

TECHNICAL FIELD

(1) The embodiments generally relate to the field of electric sign devices.

BACKGROUND

(2) Electric signs can typically include light emitting diode (LED) strips adhered to metal text characters. However, such signs are not customizable as the positions of the metal text characters are generally completely fixed.

(3) There is a need for an electric sign that includes text characters that are rearrangeable.

SUMMARY

(4) This summary is provided to introduce a variety of concepts in a simplified form that is further disclosed in the detailed description of the embodiments. This summary is not intended to identify key or essential inventive concepts of the claimed subject matter, nor is it intended for determining the scope of the claimed subject matter.

(5) In general, the disclosed electric sign device can include a backboard; first and second metal strips attached to the backboard; a plurality of LED assemblies, each LED assembly comprising: a board; first and second metal contacts attached to a first side of the board; first and second magnets attached to the first and second metal contacts, respectively; the first and second metal contacts contacting the first and second metal strips, respectively; a plurality of LEDs disposed on a second

side of the board, the plurality of LEDs being electrically connected to the first and second metal contacts; and a carrier having one or more channels or apertures that receive the plurality of LEDs, the carrier being attached to the board; wherein the first and second magnets are magnetically attached to the first and second metal strips, respectively, so that the LED assembly is magnetically attached to the backboard.

(6) Other illustrative variations within the scope of the invention will become apparent from the detailed description provided hereinafter. The detailed description and enumerated variations, while disclosing optional variations, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more complete understanding of the embodiments, and the attendant advantages and features thereof, will be more readily understood by references to the following detailed description when considered in conjunction with the accompanying drawings wherein:

(2) FIG. 1 illustrates an illustrative depiction of an electric sign, in accordance with some embodiments;

(3) FIG. 2 illustrates an illustrative depiction of an LED assembly, in accordance with some embodiments;

(4) FIG. 3 illustrates an illustrative depiction of an LED assembly with a carrier omitted, in accordance with some embodiments;

(5) FIG. 4 illustrates an illustrative depiction of a carrier of an LED assembly, in accordance with some embodiments;

(6) FIG. 5 illustrates an illustrative depiction of an LED assembly that includes a carrier cover, in accordance with some embodiments;

(7) FIG. 6 illustrates an illustrative depiction of an LED assembly that includes a carrier cover, in accordance with some embodiments; and

(8) FIG. 7 illustrates an illustrative depiction of a side view of an electric sign, in accordance with some embodiments.

(9) The drawings are not necessarily to scale, and certain features and certain views of the drawings may be shown exaggerated in scale or in schematic in the interest of clarity and conciseness.

DETAILED DESCRIPTION

(10) The specific details of the single embodiment or variety of embodiments described herein are to the described product or methods of use. Any specific details of the embodiments are used for demonstration purposes only and no unnecessary limitations or inferences are to be understood from there.

(11) It is noted that the embodiments reside primarily in combinations of components and procedures related to the products. Accordingly, the product and components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

(12) In general, the embodiments described herein relate to an electric sign device that can include a backboard. The electric sign device can include first and second metal strips attached to the backboard and a plurality of LED assemblies, wherein each LED assembly comprises: a board; first and second metal contacts attached to a first side of the board; first and second magnets attached to the first and second metal contacts, respectively; the first and second metal contacts contacting the first and second metal strips, respectively; a plurality of LEDs disposed on a second

side of the board, the plurality of LEDs being electrically connected to the first and second metal contacts; and a carrier having one or more channels or apertures that receive the plurality of LEDs, the carrier being attached to the board; wherein the first and second magnets are magnetically attached to the first and second metal strips, respectively, so that the each LED assembly is magnetically attached to the backboard.

(13) In some embodiments, the backboard and carrier can be made primarily of acrylic. In some embodiments, the board of each LED assembly can be a printed circuit board (PCB) such as, for example, a FR-4 fiberglass PCB having a thickness of approximately 0.03 inches. The first and second metal strips can be made primarily of galvanized steel. In some embodiments, the first and second magnets can comprise neodymium and can be approximately 7 mm in diameter and 2 mm in thickness. Any electronics can be attached to a board of an LED assembly by soldering, using for example, a pick-and-place machine and a surface mount technology reflow oven. In some embodiments, the carrier and board can be cut or shaped by using computer numerical control (CNC) machining and/or laser cutting.

(14) Referring to FIG. 1, an electric sign device **10** can include a first board **12** (e.g., a backboard). A first metal strip **30** can be attached to the first board **12**, the first metal strip **30** having at least a first length **40** and extending along the first board **12**. A second metal strip **32** can be attached to the first board **12**, the second metal strip **32** having at least the first length **40** and extending along the first board **12**, the second metal strip **32** being separated from the first metal strip **30** by a first predetermined distance **42**. In some embodiments, the first metal strip **30** extends in a first direction **46**, and the second metal strip **32** extends in a second direction **48** that is parallel to the first direction **46**.

(15) The electric sign device **10** can include one or more light emitting diode (LED) assemblies **20**. The one or more LED assemblies **20** can include LED assemblies **21**, **22**, **23**, **24**, **25**, **26**, and **27**. In some embodiments, the LED assemblies **21**, **22**, **23**, etc. can be formed to represent respective text characters “H”, “A”, “P”, “P”, “Y”, B, “I”, “R”, “T”, “H”, “D”, “A”, “Y”, representing a phrase or message “HAPPY BIRTHDAY.” However, in other embodiments, any suitable text characters can be represented by the LED assemblies **21**, **22**, **23**, etc. For example, any English letters, symbols (e.g., mathematical symbols), punctuation marks/symbols, non-English characters, emojis, etc. can be represented by the LED assemblies **21**, **22**, **23**, etc. In some embodiments, the text characters represented by the LED assemblies **21**, **22**, **23**, etc. can be selected to represent any suitable phrase or message.

(16) In some embodiments, any suitable number of LED assemblies can be included in the one or more LED assemblies **20**. In some embodiments, the first metal strip **30** or the second metal strip **32** can be constructed and arranged to be connected to a power source **120** by an electrical connection **124** to provide power to the LEDs in the LED assemblies **20**. In some embodiments, the first metal strip **30** or the second metal strip **32** can be constructed and arranged to be connected to a reference potential **122** (e.g., ground) by an electrical connection **124**.

(17) Referring to FIG. 2, an LED assembly **27** can include a second board **50** having a first width **60** that is smaller than the first length (e.g., **40** in FIG. 1). In some embodiments, the LED assembly **27** can include a first magnet **54**. The LED assembly **27** can include a first metal contact **56** attached to a first side **51** of the second board **50**. In some embodiments, the first metal contact **56** can be constructed and arranged to receive the first magnet **54**. In some embodiments, the first magnet **54** can be attached (e.g., magnetically attached) to the first metal contact **56**. In some embodiments, the first magnet **54** can be attached to the first side **51** of the second board **50**, wherein the first magnet **54** is proximate to the first metal contact **56**.

(18) In some embodiments, the LED assembly **27** can include a second magnet **64**. A second metal contact **66** can be attached to the first side **51** of the second board **50**, the second metal contact **66** being separated from the first metal contact **56** by a second predetermined distance **44**. The second metal contact **66** can be constructed and arranged to receive the second magnet **64**. In some

embodiments, the second magnet **64** can be separated from the first magnet **54**. In some embodiments, the second magnet **64** can be attached (e.g., magnetically attached) to the second metal contact **66**. In some embodiments, the second magnet **64** can be attached to the first side **51** of the second board **50**, wherein the second magnet **64** is proximate to the second metal contact **66**. (19) While two metal contacts **56**, **66** are illustrated, any suitable number of metal contacts can be included in the LED assembly **27**. While two magnets **54**, **64** are illustrated, any suitable number of magnets can be included in the LED assembly **27**.

(20) Referring to FIG. **3**, a first plurality of LEDs **70** can be disposed on a second side **52** of the second board **50**, the first plurality of LEDs **70** being electrically connected to the first metal contact (e.g., **56** in FIG. **2**) and the second metal contact (e.g., **66** in FIG. **2**). The first plurality of LEDs **70** can be soldered on the second board **50**. Electrical connections **74** can be formed between the LEDs **70**. In some embodiments, one or more resistors **76** can be electrically connected between one or more LEDs of the plurality of LEDs **70**. Electrical vias **75** can electrically connect the plurality of LEDs **70** on the second side **52** of the second board **50** to the first metal contact and the second metal contact attached to the first side **51** of the second board **50**.

(21) In some embodiments, the plurality of LEDs **70** can include at least a first LED **71**, a second LED **72**, and a third LED **73**. In some embodiments, the LEDs **71**, **72**, **73** of the plurality of LEDs **70** are electrically connected in series. Any suitable number of LEDs can be included in the plurality of LEDs **70**.

(22) Referring to FIG. **4**, a first carrier **80** can have one or more first channels or apertures **82** that are constructed and arranged to at least partially receive the first plurality of LEDs (e.g., **70** in FIG. **3**). After receiving the first plurality of LEDs, a fluid seal can be formed between the first carrier **80** and the second board **50** to at least partially fluidly seal the one or more channels or apertures and the LEDs. In some embodiments, the fluid seal can be formed by using epoxy (e.g., epoxy resin). Referring to FIGS. **2** and **4**, the first carrier **80** can have one or more channels **84** that are constructed and arranged to receive the second board **50**. The second board **50** can be attached to the first carrier **80** and can be positioned at least partially inside the one or more channels **84**. In some embodiments, another fluid seal can be formed using epoxy (e.g., epoxy resin) on the first side **51** of the second board **50** and the first carrier **80** to fluidly seal the second board **50** to the first carrier **80**. In some embodiments, the first carrier **80** can be at least partially transparent.

(23) Referring to FIG. **5**, in some embodiments, an LED assembly **26** can include a carrier cover **110** attached to a carrier **80'**. The carrier cover **110** can be constructed and arranged to cover at least a portion **81** of the carrier **80'**. In some embodiments, the carrier cover **110** can be at least partially transparent. In some embodiments, the carrier cover **110** can be less transparent than the carrier **80'** to, for example, diffuse more light from the LEDs **70'** than the carrier **80'**. The carrier cover **110** can be positioned over the plurality of LEDs **70'**. The plurality of LEDs **70'** can be disposed on a second side **52'** of a second board **50'**. The LEDs **70'** can be positioned at least partially inside the one or more channels or apertures **82'**. The plurality of LEDs **70'** can be positioned at least partially between the second side **52'** and the carrier cover **110**. In some embodiments, the one or more channels or apertures **82'** do not extend through (e.g., not entirely through) the carrier **80'**. In some embodiments, the carrier cover **110** can be made primarily of acrylic.

(24) Referring to FIG. **6**, an LED assembly **26** can include a carrier cover **110'** that is attached to a carrier **80''** by being engaged with the carrier **80''**. For example, one or more grooves **93** in the carrier **80''** can receive a protrusion **95** of the carrier cover **110'** so that the carrier cover **110'** is retained by the carrier **80''**. As another example, one or more grooves **96** of the carrier cover **110'** can receive a protrusion **94** of the carrier **80''** so that the carrier cover **110'** is retained by the carrier **80''**. In some embodiments, the carrier cover **110'** can be made primarily of silicone. The LED assembly **26** can have one or more channels or apertures **82''** that are constructed and arranged to receive the plurality of LEDs **70''**. The LEDs **70''** can be positioned at least partially inside the one or more channels or apertures **82''**. In some embodiments, the one or more channels or apertures

82" can extend through (e.g., entirely through) the carrier 80".

(25) Referring to FIG. 7, in some embodiments, the first magnet 54 is constructed and arranged to be magnetically attached to the first metal strip 30 and the second magnet 64 is constructed and arranged to be magnetically attached to the second metal strip 32 so that the LED assembly 27 is magnetically attached to the first board 12. In some embodiments, the first magnet 54 and the second magnet 64 are constructed and arranged to be simultaneously magnetically attached to the first metal strip 30 and the second metal strip 32, respectively.

(26) In some embodiments, the first board 12 can have a first groove 90 that is constructed and arranged to receive the first metal strip 30. The first metal strip 30 can be positioned inside the first groove 90. At least a portion 106 of the first metal contact 56 is constructed and arranged to be positioned inside the first groove 90. In some embodiments, the first board 12 can have a second groove 92 that is constructed and arranged to receive the second metal strip 32. The second metal strip 32 can be positioned inside the second groove 92. At least a portion 116 of the second metal contact 66 is constructed and arranged to be positioned inside the second groove 92. The first metal contact 56 and the second metal contact 66 can abut respective edges 97 defining the first groove 90 and the second groove 92 so that the LED assembly 27 is prevented from sliding in a downward direction 99.

(27) In some embodiments, the first metal contact 56 can be constructed and arranged to electrically connect to (e.g., contact) the first metal strip 30. In some embodiments, the second metal contact 66 can be constructed and arranged to electrically connect to (e.g., contact) the second metal strip 32. In some embodiments, the first metal contact 56 and the second metal contact 66 are constructed and arranged to simultaneously electrically connect to (e.g., contact) the first metal strip 30 and the second metal strip 32, respectively. In some embodiments, the second predetermined distance 44 can be approximately the same as the first predetermined distance 42. In some embodiments, the second predetermined distance 44 can be smaller than the first predetermined distance 42, if for example the first metal contact 56 and the second metal contact 66 are sufficiently large to simultaneously electrically connect to the first metal strip 30 and the second metal strip 32, respectively. In some embodiments, the second predetermined distance 44 can be larger than the first predetermined distance 42, if for example the first metal contact 56 and the second metal contact 66 are sufficiently small to simultaneously electrically connect to the first metal strip 30 and the second metal strip 32, respectively. The LED assembly 27 can be constructed and arranged to electrically connect the first metal strip 30 to the second metal strip 32 while the LED assembly 27 is magnetically attached to the first board 12.

(28) Referring to FIG. 1, any LED assembly 21, 22, 23, etc. of the one or more LED assemblies 20 is detachable from the first metal strip 30 and the second metal strip 32. After being detached, any LED assembly can be reattached at any position along the length 40 of the first metal strip 30 and the second metal strip 32. Any LED assembly 21, 22, 23, etc. can be rearranged from any position along the length 40 to any other position along the length 40 of the first metal strip 30 and the second metal strip 32. In some embodiments, the one or more LED assemblies 20 can be rearranged along the length of the first metal strip 30 and second metal strip 32 to convey different messages or phrases. In some embodiments, the one or more LED assemblies 20 are rearrangeable along the length of the first metal strip 30 and second metal strip 32.

(29) The following description of variants is only illustrative of components, elements, acts, products, and methods considered to be within the scope of the invention and are not in any way intended to limit such scope by what is specifically disclosed or not expressly set forth. The components, elements, acts, products, and methods as described herein may be combined and rearranged other than as expressly described herein and are still considered to be within the scope of the invention.

(30) According to variation 1, an electric sign device can include a first board; a first metal strip attached to the first board, the first metal strip having at least a first length and extending along the

first board; a second metal strip attached to the first board, the second metal strip having at least the first length and extending along the first board, the second metal strip being separated from the first metal strip; a first light emitting diode (LED) assembly, comprising: a second board having a first width that is smaller than the first length; a first magnet; a first metal contact attached to a first side of the second board, the first metal contact being constructed and arranged to receive the first magnet, the first metal contact being constructed and arranged to contact the first metal strip; a second magnet; a second metal contact attached to the first side of the second board, the second metal contact being separated from the first metal contact, the second metal contact being constructed and arranged to receive the second magnet, the second metal contact being constructed and arranged to contact the second metal strip; a first plurality of LEDs disposed on a second side of the second board, the first plurality of LEDs being electrically connected to the first metal contact and the second metal contact; and a first carrier having one or more first channels or apertures that are constructed and arranged to at least partially receive the first plurality of LEDs, the first carrier being attached to the second board; wherein the first magnet is constructed and arranged to be magnetically attached to the first metal strip and the second magnet is constructed and arranged to be magnetically attached to the second metal strip so that the first LED assembly is magnetically attached to the first board; a second LED assembly, comprising: a third board having a second width that is smaller than the first length; a third magnet; a third metal contact attached to a first side of the third board, the third metal contact being constructed and arranged to receive the third magnet, the third metal contact being constructed and arranged to contact the first metal strip; a fourth magnet; a fourth metal contact attached to the first side of the third board, the fourth metal contact being separated from the third metal contact, the fourth metal contact being constructed and arranged to receive the fourth magnet, the fourth metal contact being constructed and arranged to contact the second metal strip; a second plurality of LEDs disposed on a second side of the third board, the second plurality of LEDs being electrically connected to the third metal contact and the fourth metal contact; and a second carrier having one or more second channels or apertures that are constructed and arranged to at least partially receive the second plurality of LEDs, the second carrier being attached to the third board; wherein the third magnet is constructed and arranged to be magnetically attached to the first metal strip and the fourth magnet is constructed and arranged to be magnetically attached to the second metal strip so that the second LED assembly is magnetically attached to the first board.

(31) Variation 2 can include the electric sign device of variation 1, wherein the first carrier is formed to represent a first text character, and wherein the second carrier is formed to represent a second text character that is different than the first text character.

(32) Variation 3 can include the electric sign device of variation 1, wherein the first LED assembly further comprises: a first carrier cover attached to the first carrier, the first carrier cover being constructed and arranged to cover at least a portion of the first carrier, the first carrier cover being at least partially transparent.

(33) Variation 4 can include the electric sign device of variation 1, wherein: the first board has a first groove; the first metal strip is positioned inside the first groove; at least a portion of the first metal contact is constructed and arranged to be positioned inside the first groove; the first board has a second groove; the second metal strip is positioned inside the second groove; and at least a portion of the second metal contact is constructed and arranged to be positioned inside the second groove.

(34) Variation 5 can include the electric sign device of variation 1, wherein the first LED assembly is constructed and arranged to electrically connect the first metal strip to the second metal strip while the first LED assembly is magnetically attached to the first board.

(35) Variation 6 can include the electric sign device of variation 1, wherein the first or second metal strip is constructed and arranged to be connected to a power source.

(36) Variation 7 can include the electric sign device of variation 1, wherein LEDs of the first

plurality of LEDs are electrically connected in series.

(37) Variation 8 can include the electric sign device of variation 1, wherein: the first magnet is a first neodymium magnet; and the second magnet is a second neodymium magnet.

(38) Variation 9 can include the electric sign device of variation 1, wherein the first metal strip extends in a first direction, and the second metal strip extends in a second direction that is parallel to the first direction.

(39) According to variation 10, an electric sign device can include a first board; a first metal strip attached to the first board, the first metal strip having at least a first length and extending along the first board; a second metal strip attached to the first board, the second metal strip having at least the first length and extending along the first board, the second metal strip being separated from the first metal strip; a first light emitting diode (LED) assembly, comprising: a second board having a first width that is smaller than the first length; a first magnet; a first metal contact attached to a first side of the second board, the first magnet being constructed and arranged to be attached to the first metal contact, the first metal contact being constructed and arranged to electrically connect to the first metal strip; a second magnet; a second metal contact attached to the first side of the second board, the second metal contact being separated from the first metal contact, the second magnet being constructed and arranged to be attached to the second metal contact, the second metal contact being constructed and arranged to electrically connect to the second metal strip; a first plurality of LEDs disposed on a second side of the second board, the first plurality of LEDs being electrically connected to the first metal contact and the second metal contact; and a first carrier having one or more first channels or apertures that are constructed and arranged to at least partially receive the first plurality of LEDs, the first carrier being attached to the second board; wherein the first magnet is constructed and arranged to be magnetically attached to the first metal strip and the second magnet is constructed and arranged to be magnetically attached to the second metal strip so that the first LED assembly is magnetically attached to the first board; a second LED assembly, comprising: a third board having a second width that is smaller than the first length; a third magnet; a third metal contact attached to a first side of the third board, the third magnet being constructed and arranged to be attached to the third metal contact, the third metal contact being constructed and arranged to electrically connect to the first metal strip; a fourth magnet; a fourth metal contact attached to the first side of the third board, the fourth metal contact being separated from the third metal contact, the fourth magnet being constructed and arranged to be attached to the fourth metal contact, the fourth metal contact being constructed and arranged to electrically connect to the second metal strip; a second plurality of LEDs disposed on a second side of the third board, the second plurality of LEDs being electrically connected to the third metal contact and the fourth metal contact; and a second carrier having one or more second channels or apertures that are constructed and arranged to at least partially receive the second plurality of LEDs, the second carrier being attached to the third board; wherein the third magnet is constructed and arranged to be magnetically attached to the first metal strip and the fourth magnet is constructed and arranged to be magnetically attached to the second metal strip so that the second LED assembly is magnetically attached to the first board.

(40) Variation 11 can include the electric sign device of variation 10, wherein the first carrier is formed to represent a first text character, and wherein the second carrier is formed to represent a second text character that is different than the first text character.

(41) Variation 12 can include the electric sign device of variation 10, wherein the first LED assembly further comprises: a first carrier cover attached to the first carrier, the first carrier cover being constructed and arranged to cover at least a portion of the first carrier, the first carrier cover being at least partially transparent.

(42) Variation 13 can include the electric sign device of variation 10, wherein: the first board has a first groove; the first metal strip is positioned inside the first groove; at least a portion of the first metal contact is constructed and arranged to be positioned inside the first groove; the first board has

a second groove; the second metal strip is positioned inside the second groove; and at least a portion of the second metal contact is constructed and arranged to be positioned inside the second groove.

(43) Variation 14 can include the electric sign device of variation 10, wherein the first LED assembly is constructed and arranged to electrically connect the first metal strip to the second metal strip while the first LED assembly is magnetically attached to the first board.

(44) Variation 15 can include the electric sign device of variation 10, wherein the first or second metal strip is constructed and arranged to be connected to a power source.

(45) Variation 16 can include the electric sign device of variation 10, wherein LEDs of the first plurality of LEDs are electrically connected in series.

(46) Variation 17 can include the electric sign device of variation 10, wherein: the first magnet is a first neodymium magnet; and the second magnet is a second neodymium magnet.

(47) Variation 18 can include the electric sign device of variation 10, wherein the first metal strip extends in a first direction, and the second metal strip extends in a second direction that is parallel to the first direction.

(48) According to variation 19, an electric sign device can include a first board; a first metal strip attached to the first board, the first metal strip having at least a first length and extending along the first board; a second metal strip attached to the first board, the second metal strip having at least the first length and extending along the first board, the second metal strip being separated from the first metal strip; a first light emitting diode (LED) assembly, comprising: a second board having a first width that is smaller than the first length; a first magnet attached to a first side of the second board; a first metal contact attached to the first side of the second board, the first magnet being constructed and arranged to be proximate to the first metal contact, the first metal contact being constructed and arranged to electrically connect to the first metal strip; a second magnet attached to the first side of the second board; a second metal contact attached to the first side of the second board, the second magnet being constructed and arranged to be proximate to the second metal contact, the second metal contact being separated from the first metal contact, the second metal contact being constructed and arranged to electrically connect to the second metal strip; a first plurality of LEDs disposed on a second side of the second board, the first plurality of LEDs being electrically connected to the first metal contact and the second metal contact; and a first carrier having one or more first channels or apertures that are constructed and arranged to at least partially receive the first plurality of LEDs, the first carrier being attached to the second board; wherein the first magnet is constructed and arranged to be magnetically attached to the first metal strip and the second magnet is constructed and arranged to be magnetically attached to the second metal strip so that the first LED assembly is magnetically attached to the first board; a second LED assembly, comprising: a third board having a second width that is smaller than the first length; a third magnet attached to a first side of the third board; a third metal contact attached to the first side of the third board, the third magnet being constructed and arranged to be proximate to the third metal contact, the third metal contact being constructed and arranged to electrically connect to the first metal strip; a fourth magnet attached to the first side of the third board; a fourth metal contact attached to the first side of the third board, the fourth metal contact being separated from the third metal contact, the fourth magnet being constructed and arranged to be proximate to the fourth metal contact, the fourth metal contact being constructed and arranged to electrically connect to the second metal strip; a second plurality of LEDs disposed on a second side of the third board, the second plurality of LEDs being electrically connected to the third metal contact and the fourth metal contact; and a second carrier having one or more second channels or apertures that are constructed and arranged to at least partially receive the second plurality of LEDs, the second carrier being attached to the third board; wherein the third magnet is constructed and arranged to be magnetically attached to the first metal strip and the fourth magnet is constructed and arranged to be magnetically attached to the second metal strip so that the second LED assembly is magnetically

attached to the first board.

(49) Variation 20 can include the electric sign device of variation 19, wherein the first carrier is formed to represent a first text character, and wherein the second carrier is formed to represent a second text character that is different than the first text character.

(50) Many different embodiments have been disclosed herein, in connection with the above description and the drawings. It will be understood that it would be unduly repetitious and obfuscating to describe and illustrate every combination and sub-combination of these embodiments. Accordingly, all embodiments can be combined in any way and/or combination, and the present specification, including the drawings, shall be construed to constitute a complete written description of all combinations and sub-combinations of the embodiments described herein, and of the manner and process of making and using them, and shall support claims to any such combination or sub-combination.

(51) An equivalent substitution of two or more elements can be made for any one of the elements in the claims below or that a single element can be substituted for two or more elements in a claim. Although elements can be described above as acting in certain combinations, and even initially claimed as such, it is to be expressly understood that one or more elements from a claimed combination can, in some cases, be excised from the combination and that the claimed combination can be directed to a sub-combination or variation of a sub-combination.

(52) It will be appreciated by persons skilled in the art that the present embodiment is not limited to what has been particularly shown and described hereinabove. A variety of modifications and variations are possible considering the above teachings without departing from the following claims.

Claims

1. An electric sign device, comprising: a first board; a first metal strip attached to the first board, the first metal strip having at least a first length and extending along the first board; a second metal strip attached to the first board, the second metal strip having at least the first length and extending along the first board, the second metal strip being separated from the first metal strip; a first light emitting diode (LED) assembly, comprising: a second board having a first width that is smaller than the first length; a first magnet; a first metal contact attached to a first side of the second board, the first metal contact being constructed and arranged to receive the first magnet, the first metal contact being constructed and arranged to contact the first metal strip; a second magnet; a second metal contact attached to the first side of the second board, the second metal contact being separated from the first metal contact, the second metal contact being constructed and arranged to receive the second magnet, the second metal contact being constructed and arranged to contact the second metal strip; a first plurality of LEDs disposed on a second side of the second board, the first plurality of LEDs being electrically connected to the first metal contact and the second metal contact; and a first carrier having one or more first channels or apertures that are constructed and arranged to at least partially receive the first plurality of LEDs, the first carrier being attached to the second board; wherein the first magnet is constructed and arranged to be magnetically attached to the first metal strip and the second magnet is constructed and arranged to be magnetically attached to the second metal strip so that the first LED assembly is magnetically attached to the first board; a second LED assembly, comprising: a third board having a second width that is smaller than the first length; a third magnet; a third metal contact attached to a first side of the third board, the third metal contact being constructed and arranged to receive the third magnet, the third metal contact being constructed and arranged to contact the first metal strip; a fourth magnet; a fourth metal contact attached to the first side of the third board, the fourth metal contact being separated from the third metal contact, the fourth metal contact being constructed and arranged to receive the fourth magnet, the fourth metal contact being constructed and arranged to contact the second metal

strip; a second plurality of LEDs disposed on a second side of the third board, the second plurality of LEDs being electrically connected to the third metal contact and the fourth metal contact; and a second carrier having one or more second channels or apertures that are constructed and arranged to at least partially receive the second plurality of LEDs, the second carrier being attached to the third board; wherein the third magnet is constructed and arranged to be magnetically attached to the first metal strip and the fourth magnet is constructed and arranged to be magnetically attached to the second metal strip so that the second LED assembly is magnetically attached to the first board.

2. The electric sign device of claim 1, wherein the first carrier is formed to represent a first text character, and wherein the second carrier is formed to represent a second text character that is different than the first text character.

3. The electric sign device of claim 1, wherein the first LED assembly further comprises: a first carrier cover attached to the first carrier, the first carrier cover being constructed and arranged to cover at least a portion of the first carrier, the first carrier cover being at least partially transparent.

4. The electric sign device of claim 1, wherein: the first board has a first groove; the first metal strip is positioned inside the first groove; at least a portion of the first metal contact is constructed and arranged to be positioned inside the first groove; the first board has a second groove; the second metal strip is positioned inside the second groove; and at least a portion of the second metal contact is constructed and arranged to be positioned inside the second groove.

5. The electric sign device of claim 1, wherein the first LED assembly is constructed and arranged to electrically connect the first metal strip to the second metal strip while the first LED assembly is magnetically attached to the first board.

6. The electric sign device of claim 1, wherein the first or second metal strip is constructed and arranged to be connected to a power source.

7. The electric sign device of claim 1, wherein LEDs of the first plurality of LEDs are electrically connected in series.

8. The electric sign device of claim 1, wherein: the first magnet is a first neodymium magnet; and the second magnet is a second neodymium magnet.

9. The electric sign device of claim 1, wherein the first metal strip extends in a first direction, and the second metal strip extends in a second direction that is parallel to the first direction.

10. An electric sign device, comprising: a first board; a first metal strip attached to the first board, the first metal strip having at least a first length and extending along the first board; a second metal strip attached to the first board, the second metal strip having at least the first length and extending along the first board, the second metal strip being separated from the first metal strip; a first light emitting diode (LED) assembly, comprising: a second board having a first width that is smaller than the first length; a first magnet; a first metal contact attached to a first side of the second board, the first magnet being constructed and arranged to be attached to the first metal contact, the first metal contact being constructed and arranged to electrically connect to the first metal strip; a second magnet; a second metal contact attached to the first side of the second board, the second metal contact being separated from the first metal contact, the second magnet being constructed and arranged to be attached to the second metal contact, the second metal contact being constructed and arranged to electrically connect to the second metal strip; a first plurality of LEDs disposed on a second side of the second board, the first plurality of LEDs being electrically connected to the first metal contact and the second metal contact; and a first carrier having one or more first channels or apertures that are constructed and arranged to at least partially receive the first plurality of LEDs, the first carrier being attached to the second board; wherein the first magnet is constructed and arranged to be magnetically attached to the first metal strip and the second magnet is constructed and arranged to be magnetically attached to the second metal strip so that the first LED assembly is magnetically attached to the first board; a second LED assembly, comprising: a third board having a second width that is smaller than the first length; a third magnet; a third metal contact attached to a first side of the third board, the third magnet being constructed and arranged

to be attached to the third metal contact, the third metal contact being constructed and arranged to electrically connect to the first metal strip; a fourth magnet; a fourth metal contact attached to the first side of the third board, the fourth metal contact being separated from the third metal contact, the fourth magnet being constructed and arranged to be attached to the fourth metal contact, the fourth metal contact being constructed and arranged to electrically connect to the second metal strip; a second plurality of LEDs disposed on a second side of the third board, the second plurality of LEDs being electrically connected to the third metal contact and the fourth metal contact; and a second carrier having one or more second channels or apertures that are constructed and arranged to at least partially receive the second plurality of LEDs, the second carrier being attached to the third board; wherein the third magnet is constructed and arranged to be magnetically attached to the first metal strip and the fourth magnet is constructed and arranged to be magnetically attached to the second metal strip so that the second LED assembly is magnetically attached to the first board.

11. The electric sign device of claim 10, wherein the first carrier is formed to represent a first text character, and wherein the second carrier is formed to represent a second text character that is different than the first text character.

12. The electric sign device of claim 10, wherein the first LED assembly further comprises: a first carrier cover attached to the first carrier, the first carrier cover being constructed and arranged to cover at least a portion of the first carrier, the first carrier cover being at least partially transparent.

13. The electric sign device of claim 10, wherein: the first board has a first groove; the first metal strip is positioned inside the first groove; at least a portion of the first metal contact is constructed and arranged to be positioned inside the first groove; the first board has a second groove; the second metal strip is positioned inside the second groove; and at least a portion of the second metal contact is constructed and arranged to be positioned inside the second groove.

14. The electric sign device of claim 10, wherein the first LED assembly is constructed and arranged to electrically connect the first metal strip to the second metal strip while the first LED assembly is magnetically attached to the first board.

15. The electric sign device of claim 10, wherein the first or second metal strip is constructed and arranged to be connected to a power source.

16. The electric sign device of claim 10, wherein LEDs of the first plurality of LEDs are electrically connected in series.

17. The electric sign device of claim 10, wherein: the first magnet is a first neodymium magnet; and the second magnet is a second neodymium magnet.

18. The electric sign device of claim 10, wherein the first metal strip extends in a first direction, and the second metal strip extends in a second direction that is parallel to the first direction.

19. An electric sign device, comprising: a first board; a first metal strip attached to the first board, the first metal strip having at least a first length and extending along the first board; a second metal strip attached to the first board, the second metal strip having at least the first length and extending along the first board, the second metal strip being separated from the first metal strip; a first light emitting diode (LED) assembly, comprising: a second board having a first width that is smaller than the first length; a first magnet attached to a first side of the second board; a first metal contact attached to the first side of the second board, the first magnet being constructed and arranged to be proximate to the first metal contact, the first metal contact being constructed and arranged to electrically connect to the first metal strip; a second magnet attached to the first side of the second board; a second metal contact attached to the first side of the second board, the second magnet being constructed and arranged to be proximate to the second metal contact, the second metal contact being separated from the first metal contact, the second metal contact being constructed and arranged to electrically connect to the second metal strip; a first plurality of LEDs disposed on a second side of the second board, the first plurality of LEDs being electrically connected to the first metal contact and the second metal contact; and a first carrier having one or more first channels or apertures that are constructed and arranged to at least partially receive the first plurality of LEDs,

the first carrier being attached to the second board; wherein the first magnet is constructed and arranged to be magnetically attached to the first metal strip and the second magnet is constructed and arranged to be magnetically attached to the second metal strip so that the first LED assembly is magnetically attached to the first board; a second LED assembly, comprising: a third board having a second width that is smaller than the first length; a third magnet attached to a first side of the third board; a third metal contact attached to the first side of the third board, the third magnet being constructed and arranged to be proximate to the third metal contact, the third metal contact being constructed and arranged to electrically connect to the first metal strip; a fourth magnet attached to the first side of the third board; a fourth metal contact attached to the first side of the third board, the fourth metal contact being separated from the third metal contact, the fourth magnet being constructed and arranged to be proximate to the fourth metal contact, the fourth metal contact being constructed and arranged to electrically connect to the second metal strip; a second plurality of LEDs disposed on a second side of the third board, the second plurality of LEDs being electrically connected to the third metal contact and the fourth metal contact; and a second carrier having one or more second channels or apertures that are constructed and arranged to at least partially receive the second plurality of LEDs, the second carrier being attached to the third board; wherein the third magnet is constructed and arranged to be magnetically attached to the first metal strip and the fourth magnet is constructed and arranged to be magnetically attached to the second metal strip so that the second LED assembly is magnetically attached to the first board.

20. The electric sign device of claim 19, wherein the first carrier is formed to represent a first text character, and wherein the second carrier is formed to represent a second text character that is different than the first text character.
